4 Supervision Agreement

New Description

Outcome	**Weight**
Explore of IT Areas	♦♦♦♦♦

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Outcome	**Weight**
Context of Research Project	♦♦♦♦♦

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Outcome	**Weight**
Communication of Research Outcomes	♦♦♦♦♦

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Outcome	**Weight**
Validation of Objectives and Methodology	♦♦♦♦♦

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Outcome	**Weight**
Application of Skills and Knowledge	♦♦♦♦♦

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Date	**Author**	**Comment**
2025/07/17 22:46	N/a Nikhil	Working On It
2025/07/21 10:23	N/a Nikhil	Ready to Mark
2025/07/25 09:01	Ananda Maiti	Complete

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RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Supervision Agreement

Submitted By:
N/a NIKHIL
s224234862
2025/07/21 10:23

Tutor:
Ananda MAITI

Outcome	**Weight**
Explore of IT Areas	♦♦♦♦♦
Context of Research Project	♦♦♦♦♦
Communication of Research Outcomes	♦♦♦♦♦
Validation of Objectives and Methodology	♦♦♦♦♦
Application of Skills and Knowledge	♦♦♦♦♦

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July 21, 2025

Supervisor Agreement – SIT723/SIT792

Student Name:	Nikhil
Supervisors' Names:	Dr. Ananda Maiti
Project Title:	Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation in MAVROS-Gazebo Simulations
Student Details	
Course:	SIT 723 - Research Training and Project
Research Experience:	1. Conducted academic research on IoT security and sensor networks during undergraduate studies. 2. Authored a technical report on GPS spoofing vulnerabilities in autonomous systems.
Technical Skills:	1. IoT Development: GPS sensor integration, real-time data processing (School Bus Monitoring System). 2. Robotics & Simulation: MAVROS, Gazebo, ROS (Robot Operating System). 3. Machine Learning: Pattern recognition models for anomaly detection. 4. Programming: Python, C++, ROS, embedded systems. 5. Cybersecurity: Basic penetration testing, sensor spoofing concepts.
Research Units related Information	
SIT723/SIT792 Target Grade:	High Distinction (HD)

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Expectations from the unit:	1. Develop a robust simulation environment for GPS spoofing detection in MAVROS-Gazebo. 2. Publish findings in a conference or journal (if results are significant). 3. Gain hands-on experience in cyberphysical security research. 4. Improve academic writing & research methodology skills.
When will you meet with the supervisor team?	Frequency: Weekly (or bi-weekly, depending on progress). Mode: Hybrid (In-person + Microsoft Teams). Preferred Time: Every Wednesday 4pm.
Will you continue to the Research Project units (SIT724/SIT746)?	Yes **If yes, what trimester are you planning to enrol in?** Trimester 1 2026 **Will your lead supervisor be available to supervise your research project in that trimester?** Yes **If no, what supervision arrangement have you planned so that you can complete the research project you started in this trimester?** N/A (Supervisor is available.)